MTRC

MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

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Subject & Subject Code: Managerial Economics and Financial Accounting

Semester: II **Session: 2022-23 (Even Sem.)**

Branch: B.Tech(B) **Batch: 2022-2026 (B)**

Unit No.: 3 **Date of Submission:12/04/2023**

Lecture No.	Topics Covered	Page No.	Total Page
12	THEORY OF PRODUCTION- PRODUCTION FUNCTION	63-66	4
13	LAW OF VARIABLE PROPORTIONS, LAWS OF RETURNS TO SCALE PART -I	67-69	3
14	LAW OF VARIABLE PROPORTIONS, LAWS OF RETURNS TO SCALE PART -II	70-75	6
15	LEAST COST COMBINATION OF INPUTS &ISO-PRODUCT CURVES	76-82	7
16	FIXED AND VARIABLE COST, OPPORTUNITY COST, SUNK COSTS	83-90	8
17	COST CONCEPTS- COST CURVES, COST AND OUTPUT DECISIONS, COST ESTIMATION, COST FUNCTION	91-96	6

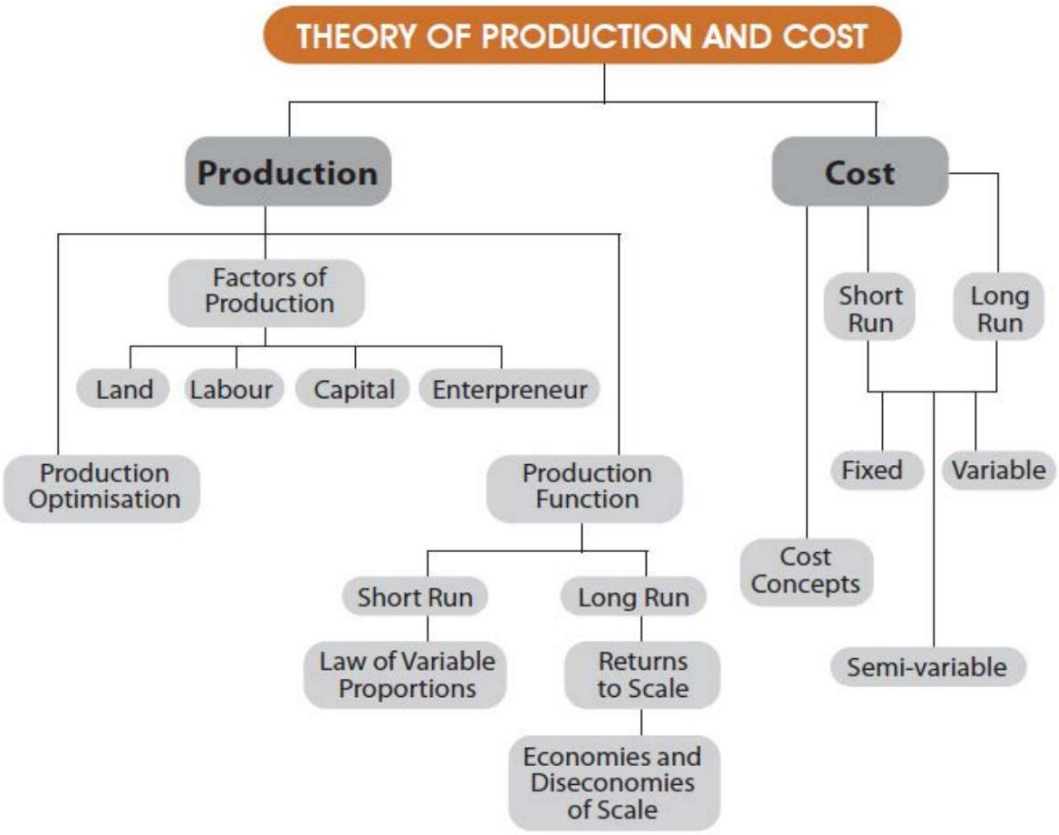
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Faculty **HOD** 12/4/2023 12/4/23 **Dean Academics** 12/4/2023

LECTURE 12

THEORY OF PRODUCTION AND COST



MEANING OF PRODUCTION

Production is a very important economic activity. As we are aware, the survival of any firm in a competitive market depends upon its ability to produce goods and services at a competitive cost. One of the principal concerns of business managers is the achievement of optimum efficiency in production by minimizing the cost of production. The performance of an economy is judged by the level of its production. The amount of goods and services an economy is able to produce determines the richness or poverty of that economy. In fact, the standard of living of people depends on the volume and variety of goods and services produced in a country. Thus, the U.S.A. is a rich country just because its level of production is high.

In common parlance, the term 'production' is used to indicate an activity of making something material. The growing of wheat, rice or any other agricultural crop by farmers

and manufacturing of cement, radio-sets, wool, machinery or any other industrial product is often referred to as production.

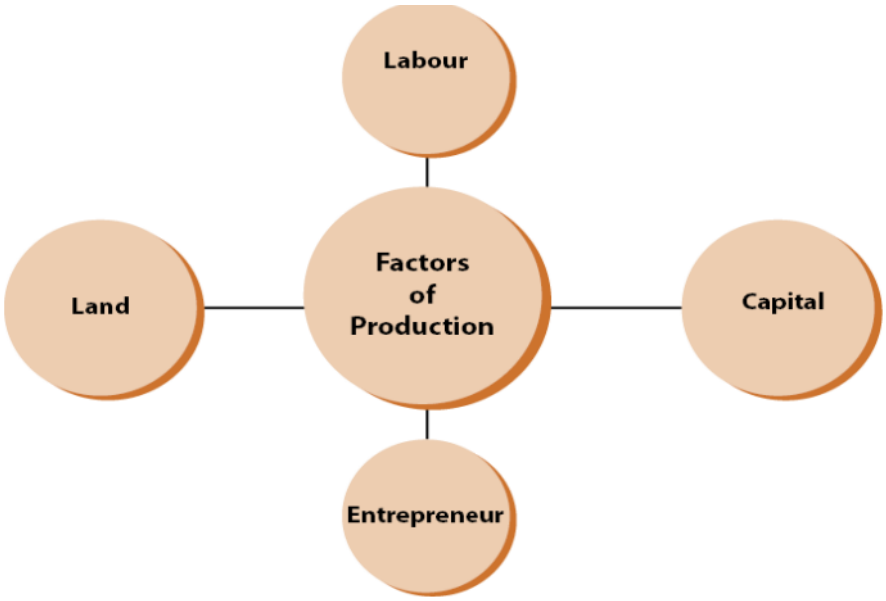
What exactly do we mean by production in Economics?

In Economics the word 'production' is used in a wider sense to denote the process by which man utilizes resources such as men, material, capital, time etc, working upon them to transform them into commodities and services so as to make them satisfy human wants. In other words, production is any economic activity which converts inputs into outputs which are capable of satisfying human wants. Whether it is making of material goods or providing a service, it is included in production provided it satisfies the wants of some people. Therefore, in Economics, activities such as making of cloth by an industrial worker, the services of the retailer who delivers it to consumers, the work of doctors, lawyers, teachers, actors, dancers, etc. are production.

According to James Bates and J.R. Parkinson "Production is the organized activity of transforming resources into finished products in the form of goods and services; and the objective of production is to satisfy the demand of such transformed resources".

FACTORS OF PRODUCTION

Factors of production refer to inputs. An input is a good or service which a firm buys for use in its production process. Production process requires a wide variety of inputs, depending on the nature of output. The process of producing goods in a modern economy is very complex. A good has to pass through many stages and many hands until it reaches the consumers' hands in a finished form. Land, labour, capital and entrepreneurial ability are the four factors or resources which make it possible to produce goods and services. Even a small piece of bread cannot be produced without the active participation of these factors of production. While land is a free gift of nature and refers to natural resources, the human endeavour is classified functionally and qualitatively into three main components namely, labour, capital and entrepreneurial skills.



- **Land** includes all those free natural resources whose supply for the economy as a whole is fixed.
- **Labour(Men)** is all human efforts of body or of mind undergone partly or wholly with a view to secure an income apart from the pleasure derived directly from the work
- **Capital(Money)** is that part of wealth of an individual or community which is used for further production of wealth. Capital, a stock concept, refers to produced means of production and it comprises of manmade machines and materials which are used for further production.
- **Entrepreneur** is the person who organises business; initiates production, remunerates other factors of production, introduces innovations and bears the risk and uncertainties of business.

Factors of production

Factors of production can be divided into two categories – Fixed factors are those factors whose quantity remains unchanged with changes in output within a capacity and variable factors are those the quantity of which change with a change in the level of output.

INPUT AND OUTPUT: -

Input means factors of production such as 5 M's Money(capital), Manpower (Labour), Methods, material & Machinery

Output means production (Commodity to be produced). In other words, an output is any commodity that comes out of production process.

FIXED FACTORS AND VARIABLE FACTORS: -

FIXED FACTOR (FF)

- Fixed factors refer to those factors of production which cannot be changed during short run.
- These are used in a fixed quantity in the short run.
- These factors can be changed only in the long run.
- Example- land, plant and machinery, factory building etc.

VARIABLE FACTOR (VF)

- Variable factor refers to those factors of production which can be changed during short period.
- The quantity of variable inputs varies according to the level of output.
- Example- labour, raw material etc.

SHORT RUN AND LONG RUN: -

- ❖ Short run: Short run refers to a period of time in which a firm cannot change its fixed factors of production, only variable factors can be changed.
- ❖ Long run: Long run refers to a time period during which a firm can change all the factors of production. In the long run, all inputs are variable. Therefore the distinction between fixed factors and variable factors will disappear.

Production function

Production function is the technical relationship between inputs and output. Samuelson describes production function as the relationship between the maximum amount of output that can be produced and the input required to make that output. It is defined for a given state of technology.

1. Law of variable proportion

The law of variable proportion or the law of diminishing returns is relevant when some factors are kept fixed and others are varied. It is applicable to the short-run. There are three stages of the law of variable proportion – where

- law of increasing returns,
- law of diminishing returns and
- law of negative returns operate

2. The Law of returns to scale

The Law of returns to scale describes the relationship between inputs and output in the long run when all inputs are changed in the same proportion. Returns to scale may be constant, increasing and decreasing.

- Constant returns to scale occur when the inputs increase by some proportion and the output also increases by the same proportion. It is also called linear homogeneous production function.

- Increasing returns to scale occur when the inputs increase by some proportion and the output increases more than proportionately.
- Decreasing returns to scale occur when the inputs increase by some proportion and the output increases less than proportionately.

3. Production optimization

Normally, a profit maximizing firm is interested to know what combination of factors of production (or inputs) would minimize its cost of production for a given output. This can be known by combining the firm's production and cost functions, namely isoquants and iso-cost lines respectively.

Production function can be stated as

Qx = F (L, C)

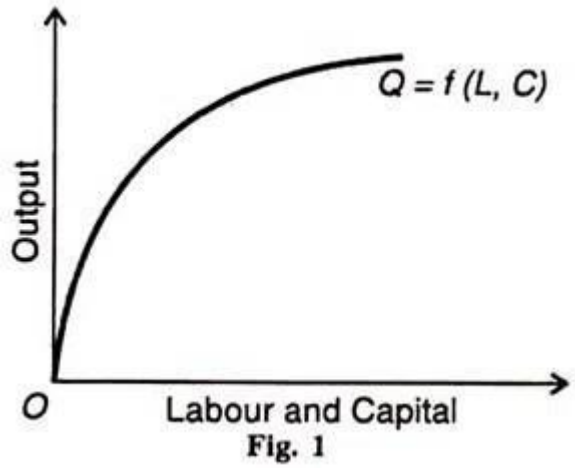


Fig. 1

Suppose there are two-factor inputs: labour (L) and capital (K). Qx is the quantity of output of commodity x, f is the function, L is the unit of labour and k is the unit capital. According to this, the amount of output depends on the labour and capital inputs utilised in manufacturing.

References:-

- Brickley. Managerial Economics & Organizational Architecture. Tata McGraw-Hill.

LECTURE 13

Cobb - Douglas Production Functions: -

The Cobb-Douglas production function is based on the empirical study of the American manufacturing industry made by Paul H. Douglas and C.W. Cobb. It is a linear homogeneous production function of degree one which takes into account two inputs, labour and capital, for the entire output of the manufacturing industry.

The Cobb-Douglas production function is expressed as:

$$Q = AL^{a} C^{\beta}$$

where Q is output and L and C are inputs of labour and capital respectively. A, a and β are positive parameters where = a > O, β > O.

The equation tells that output depends directly on L and C, and that part of output which cannot be explained by L and C is explained by A which is the 'residual', often called technical change.

The production function solved by Cobb-Douglas had 1/4 contribution of capital to the increase in manufacturing industry and 3/4 of labour so that the C-D(**Cobb-Douglas)** production function is

$$Q = AL^{3/4} C^{1/4}$$

which shows constant returns to scale because the total of the values of L and C is equal to one: (3/4 + 1/4), i.e.,(a + β = 1) . The coefficient of labourer in the C-D function measures the percentage increase in (Q that would result from a 1 per cent increase in L, while holding C as constant.

Similarly, B is the percentage increase in Q that would result from a 1 per cent increase in C, while holding L as constant. The C-D production function showing constant returns to scale is depicted in Figure 20. Labour input is taken on the horizontal axis and capital on the vertical axis.

To produce 100 units of output, OC, units of capital and OL units of labour are used. If the output were to be doubled to 200, the inputs of labour and capital would have to be doubled. OC is exactly double of OC_1 and of OL_2 is double of OL_2.

Criticisms of C-D(Cobb-Douglas) Production Function:

The C-D production function has been criticised by Arrow, Chenery, Minhas and Solow as discussed below:

1. The C-D production function considers only two inputs, labour and capital, and neglects some important inputs, like raw materials, which are used in production. It is, therefore, not possible to generalize this function to more than two inputs.

2. In the C-D production function, the problem of measurement of capital arises because it takes only the quantity of capital available for production. But the full use of the available capital can be made only in periods of full employment. This is unrealistic because no economy is always fully employed.

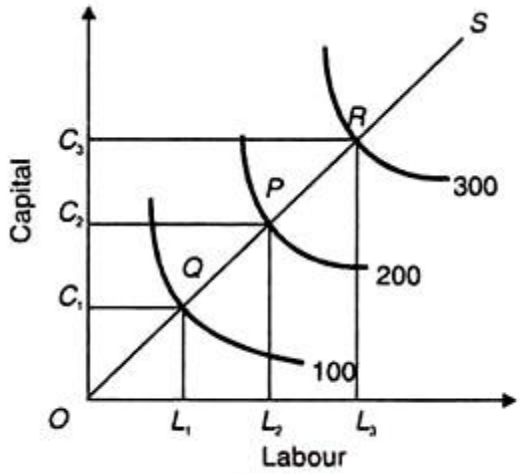


Fig. 20

3. The C-D production function is criticised because it shows constant returns to scale. But constant returns to scale are not an actuality, for either increasing or decreasing returns to scale are applicable to production.

It is not possible to change all inputs to bring a proportionate change in the outputs of all the industries.

4. The C-D production function is based on the assumption of substitutability of factors and neglects the complementarity of factors.

5. This function is based on the assumption of perfect competition in the factor market which is unrealistic. If, however, this assumption is dropped, the coefficients α and β do not represent factor shares.

6. One of the weaknesses of C-D function is the aggregation problem. This problem arises when this function is applied to every firm in an industry and to the entire industry. In this situation, there will be many production functions of low or high aggregation. Thus the C-D function does not measure what it aims at measuring.

Conclusion:

Thus the practicability of the C-D production function in the manufacturing industry is a doubtful proposition. This is not applicable to agriculture where for intensive cultivation, increasing the quantities of inputs will not raise output proportionately.

References:-

- Brickley. Managerial Economics & Organizational Architecture. Tata McGraw-Hill.

LECTURE -14

Leontief Production Function: -

Leontief production function uses fixed proportion of inputs having no substitutability between them. It is regarded as the limiting case for constant elasticity of substitution.

In economics, the **Leontief production function** or **fixed proportions production function** is a production function that implies the factors of production which will be used in fixed (technologically pre-determined) proportions, as there is no substitutability between factors. It was named after Wassily Leontief and represents a limiting case of the constant elasticity of substitution production function.

For the simple case of a good that is produced with two inputs, the function is of the form

$$q = \text{Min}\left(\frac{z_1}{a}, \frac{z_2}{b}\right)$$

where q is the quantity of output produced, z_1 and z_2 are the utilised quantities of input 1 and input 2 respectively, and a and b are technologically determined constants.

Example:-

Suppose that the intermediate goods "tires" and "steering wheels" are used in the production of automobiles (for simplicity of the example, to the exclusion of anything else). Then in the above formula q refers to the number of automobiles produced, z_1 refers to the number of tires used, and z_2 refers to the number of steering wheels used. Assuming each car is produced with 4 tires and 1 steering wheel, the Leontief production function is Number of cars = Min {1⁄4 times the number of tires, 1 time the number of steering wheels}.

References: -

1. https://www.chegg.com/learn/economics/introductiontoeconomics/demandschedule.
2. https://www.economicsdiscussion.net/demand/5typesofdemandexplained/3355.
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LECTURE -15

Law of Variable proportions- Iso-quants and Iso-costs

Laws of Variable Proportions: -

The Law of Variable Proportions is also known in the world of economics as the famous Law of Diminishing Returns. Many renowned economists and economic thinkers have opined their respective views on the concept of law of variable proportions.

Assumptions of Laws of Variable Proportions: -

The law of variable proportions valid under certain assumptions.

(i) Constant State of Technology:

First, the state of technology is assumed to be constant. In the event of improvement in the technology, the marginal product may rise instead of diminishing.

(ii) Fixed Amount of Other Factors:

Secondly, there must be some inputs whose quantity should be kept fixed.

(iii) Varying the Factor Proportions:

Thirdly, the law is based upon the prospect of varying the proportions in which the various factors can be combined to produce a product.

Illustration of the Law:

The law of variable proportion is illustrated in the following table 3.1 and figure 3.1. Let us suppose that there amount of land is given in which more and more labour (variable factor) is used to produce rice.

Table 3.1: Law of Variable Proportion

Units of Labour	Total Product	Marginal Product	Average Product
1	2	2	2
2	6	4	3
3	12	6	4
4	16	4	4
5	18	2	3.6
6	18	0	3
7	14	-4	2
8	8	-6	1

It can be seen from the table 3.1 that up to the use of 3 units of labour, total product increases at an increasing rate and beyond the third unit total product increases at a diminishing rate. This phenomenon is depicted by the marginal product which the addition is made to Total Product as a result of increasing the variable factor i.e. labour.

Production Function with Two Variable Inputs:-

Isoquants:-

A production function with two variable inputs can be represented by a tool known as isoquants.

An Isoquant is a combination of two terms, namely, iso and quant. The meaning of 'Iso' is equal. The meaning of 'Quant' is quantity. Therefore, isoquant means equal quantity or equal product. Isoquants are defined as the curves which represent the different combination of inputs producing a particular quantity of output. Any point on the isoquant corresponds to the same level of output.

Assumptions of Isoquants:

1. There are only two inputs, labour and capital to produce a commodity.

2. The two inputs (L, K) can be substituted for one another at a diminishing rate, up to a certain limit.

3. Production function is continuous.

The isoquant illustrates all possible combinations of the two inputs (say labour and capital) capable of producing equal or a given level of output. Isoquants are also known as iso product curves or equal product curves or production indifferent curves. The following example in table 3.2 explains the concept of an isoquant.

Table 3.2

Combination Units of Capital and Total Output

Combinations	Units of labour	Units of capital	Total Output
A	20	1	100
B	15	2	100
C	11	3	100
D	8	4	100
E	6	5	100

In the above schedule of table 3.2, there are five possible combinations. All the five combinations yield the same level of output i.e. 100 units. 20 units of labour and 1 unit of capital produce 100 units. 15 units of labour and 2 units of capital also produce 100 units and so on. All the combinations are equally likely because all of them produce the same level of output i.e. 100 units.

Iso-Cost Curve:-

An iso-cost line reflects the various combinations of the two factors which the firm can buy at given prices with a given amount of money. Iso-cost line depicts all the combinations of labour and capital that the firm can purchase with a given outlay and at given prices. Thus iso-cost shows the prices of the two factors and the total amount of money to spend.

Let us suppose that a firm decides to spend Rs. 500 on 2 factors – capital and labour. If the weekly wage of a worker is Rs. 50, the firm can employ 10 workers. Similarly if one unit of capital costs Rs. 20, the firm can buy 25 units of capital. Thus the firm can spend the whole amount of Rs. 500 either on labour (10 workers) or on capital (25 units) or partly on labour and partly on capital. The iso-cost line is shown in the figure 3.7.

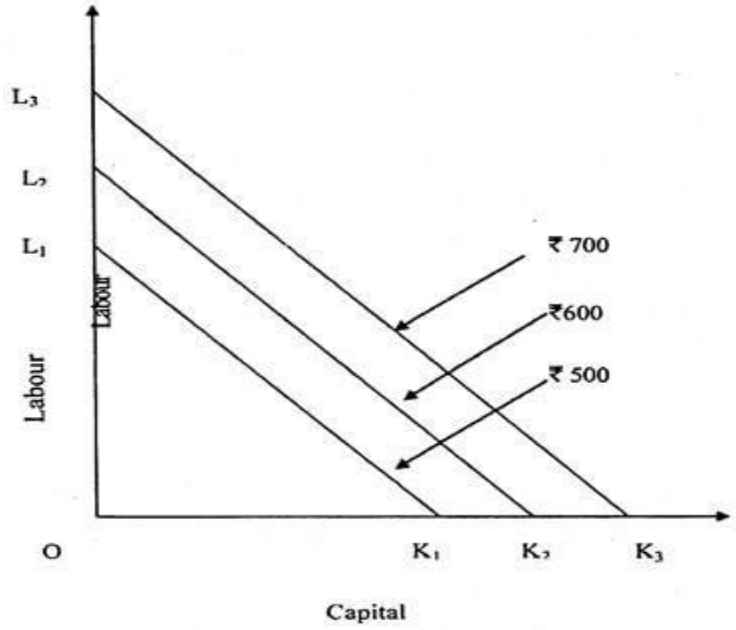


Figure 3.7 : The isocost line

The iso-cost line L_1, K_1 indicates on outlay of Rs.500. With Rs.500 the firm can buy either OL, units of labour or OK, units of capital or any combination of labour and capital between the extremes L (K., Similarly the iso-cost line L_2 K_2… shows an outlay of Rs.600 which means that either 12 workers may be employed or 30 units of capital may be bought or some units of both capital and labour. Thus iso-cost line depicts all those possible combinations of capital and labour which the firm can use with the given amount of money.

References:-

1. https://www.chegg.com/learn/economics/introductiontoeconomics/demandschedule.
2. https://www.economicsdiscussion.net/demand/5typesofdemandexplained/3355.
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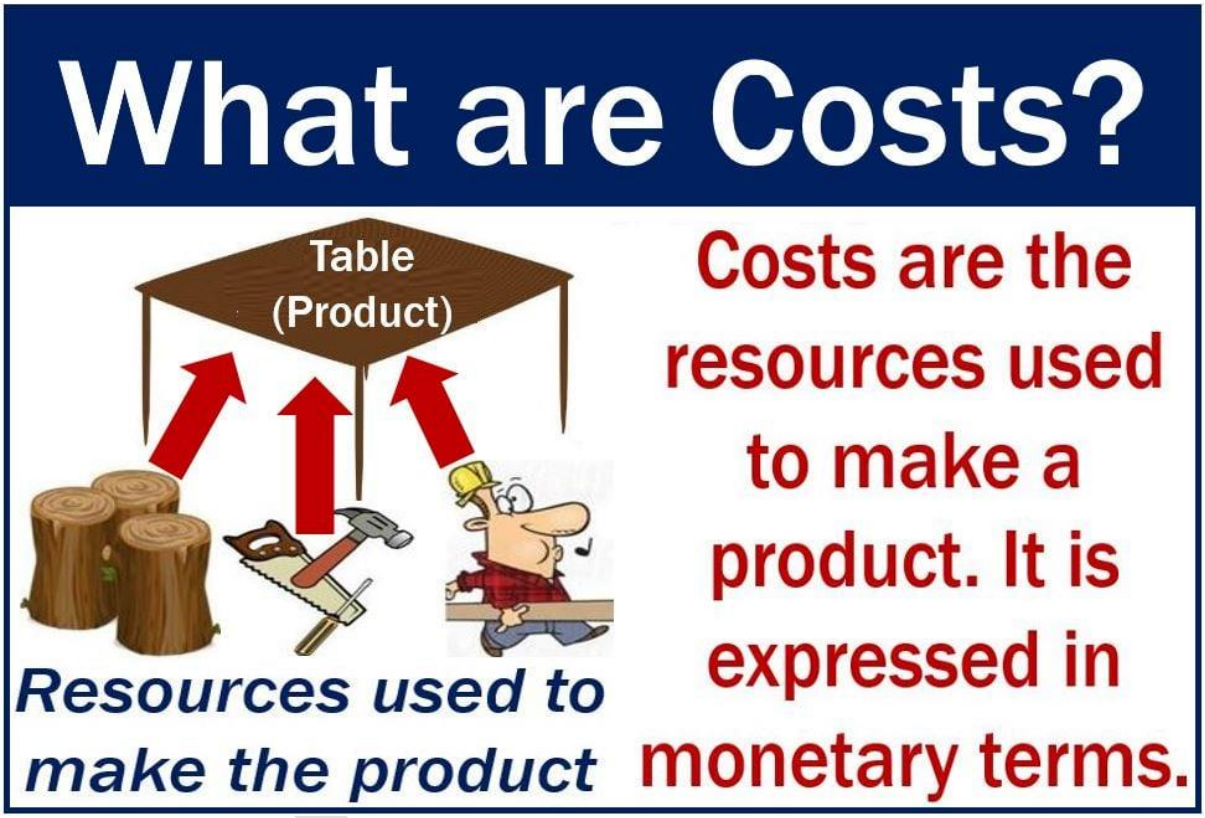
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LECTURE -16

Different Cost Concepts

Concept of Cost in Economics: -

The concept of cost is a key concept in Economics. It refers to the amount of payment made to acquire any goods and services. In a simpler way, the concept of cost is a financial valuation of resources, materials, risks, time and utilities consumed to purchase goods and services. From an economist's point of view, the cost of manufacturing any goods and services is often said to be the concept of opportunity cost.



Types of Cost Concept: -

The idea behind the concept of opportunity cost is that the cost of one item is the lost opportunity to do something else. For example, by being married to a person, one could lose the opportunity to marry some other person or by investing more capital in video games, one might lose the opportunity in watching movies.

Types of Cost Concept: -

The different types of cost concepts are:

1. Outlay costs and Opportunity costs
2. Accounting costs and Economic costs
3. Direct/Traceable costs and Indirect/Untraceable costs
4. Incremental costs and Sunk costs
5. Private costs and social costs
6. Fixed costs and Variable costs

Based on the Nature of Expenses

On the basis of nature, the following are the two types of cost:

- **Outlay Costs**

  The authentic payments undergone by an entrepreneur in employing input are known as outlay costs. It includes costs on payments of fuel, rent, electricity, etc.

- **Concept of Opportunity Cost**

  It is the value of the next best thing you give up whenever a decision is made by you.

Classification in Terms of Traceability

On the basis of traceability, the types of costs are:

1. **Direct Costs**

   A direct cost is a cost that is related to the production method of a good or service. It is the opposite of an indirect cost.

2. **Indirect Costs**

Indirect costs are expenses that could not be traced back to a single cost object or cost source. They are also known as untraceable costs. However, they are extremely important as they affect the total profitability.

Concept of Costs in Terms of Treatment

1. **Accounting Costs**

   Accounting costs are direct costs. They are also known as hard costs. The entrepreneur pays the cash directly for obtaining resources for production. It includes the cost of prices that are paid for the machines and raw materials, electricity bills, etc. These costs are treated as expenses.

2. **Economic Costs**

   The economic cost is the combination of gains and losses of the products. This cost is mainly used by economists to compare one with another.

Classification based on the Purpose

1. **Incremental Cost**

   Incremental costs are the changes in future costs and that will occur as a result after a decision is made.

2. **Sunk Costs**

   Sunk costs are the costs that cannot be recovered after sustaining. It includes the amount spent on conducting research and advertising.

Types of Cost Concept based on Players and Variability

1. **Based on Payers**

   Private cost implies the cost that is sustained when an individual produces or consumes something. The business person spends his/ her own private or business interests. The social cost is the cost to an entire society that results from a news event or a change in the policies.

2. In Terms of Variability

As the term predicts, fixed costs don't change in the volume of output. These costs are constant even with an increase or decrease in the volume of services/ goods produced or sold.

Reference: -

1.Brickley. Managerial Economics & Organizational Architecture. Tata McGraw-Hill.

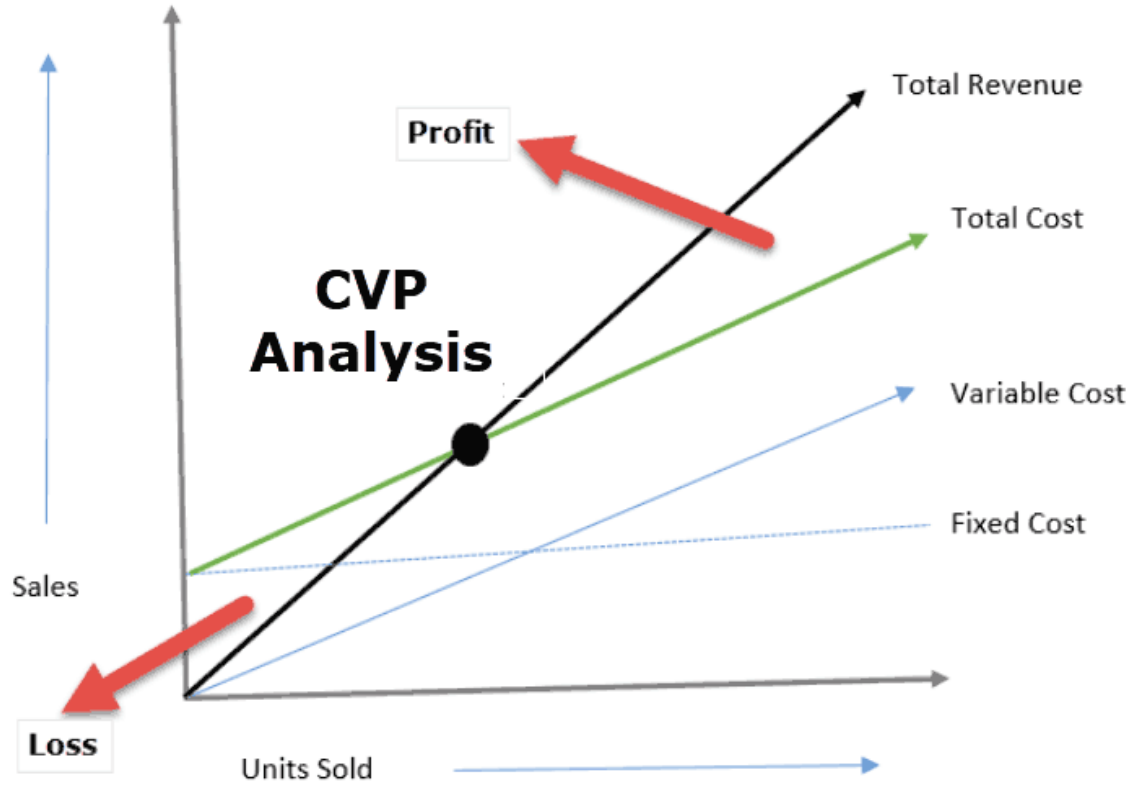
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LECTURE -17

Cost Volume Profit Analysis

Definition of Cost Volume Profit Analysis (CVP Analysis)

Cost Volume Profit Analysis (CVP) looks at the impact on the operating profit due to the varying levels of volume and the costs and determines a break-even point for cost structures with different sales volumes that will help managers in making economic decisions for short term.



Concept of Cost Volume Profit (CVP) Analysis: -

- Cost Volume Profit Analysis includes the analysis of sales price, fixed costs, variable costs, the number of goods sold, and how it affects the profit of the business.
- The aim of a company is to earn a profit, and profit depends upon a large number of factors, most notable among them is the cost of manufacturing and the volume of sales. These factors are largely interdependent.
- CVP analysis helps management in finding out the relationship between cost and revenue to generate profit.
- CVP Analysis helps them to **BEP Formula** for different sales volume and cost structures.

Cost Volume Profit Analysis Formula

The computing of Cost volume profit analysis formula is as follows:



Example

ABC Limited has entered into the business of making Electrical fans. The management of the company is interested in knowing the breakeven point at which there will be no profit/loss. Below are the details pertaining to the cost incurred:

Total Sales Revenue of the company (@$300 per fan)	$300,000
Variable Cost	$240,000
Fixed Cost	$60,000
Maximum Capacity	14000 units

No. of units sold by ABC limited: ($300000/$300) = 1000 units

Variable cost per unit= ($240000/1000) =$240

- Contribution per unit= Selling price per unit-Variable cost per unit
- = ($300-$240)
- = $60 per unit

Break-Even Point= (Fixed Cost/Contribution per unit)

- = ($60000/$60)
- =10000 units

Thus ABC limited the need to sell 10000 units of electric fans to break even at the current cost structure.

Benefits of CVP Analysis: -

1. CVP analysis provides a clear and simple understanding of the level of sales that are required for a business to break even (No profit, No loss), level of sales required to achieve targeted profit.
2. CVP analysis helps management to understand the different costs at different levels of production/sales volume. CVP analysis helps decision-makers in forecasting cost and profit on account of change in volume.
3. CVP Analysis helps businesses analyze during recessionary times the comparative effects of shutting down a business or continuing business at a loss, as it clearly bifurcates the Direct and **Indirect cost**.
4. The effects of changes in fixed and **variable cost** help management decide the optimum level of production.

Reference: -

4. Brickley. Managerial Economics & Organizational Architecture. Tata McGraw-Hill.

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